

Cell Biology & Biochemistry

Topics Covered

Biochemistry

Biomolecules and Molecular Organization
Biological Macromolecules:

- Carbohydrates
- Lipids
- Proteins
- Nucleic acids and Nucleotides
- Identification of Biomolecules (Practical)

Cell Biology

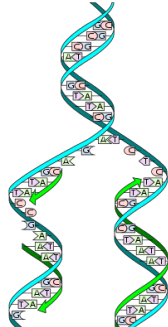
- Cell ultra-structure
- Cell Division
- Biological membranes: Structure and functions
- DNA Replication
- Protein Biosynthesis
- Cell signaling
- Bioenergetics
 - Enzymes: General properties and mechanism of action
 - Cellular respiration

BIO 1201
Cell Biology & Biochemistry
Lesson 9 - DNA Replication

Cell Biology

Biomolecules & Molecular organization
DNA replication: Overview

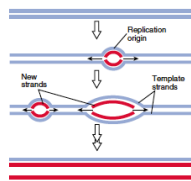
- Biological process of producing two identical copies of DNA from one original DNA molecule
- Each strand of the original DNA molecule serves as a **template** for the production of its counterpart - **semiconservative replication**



Cell Biology

Biomolecules & Molecular organization
DNA replication: Overview

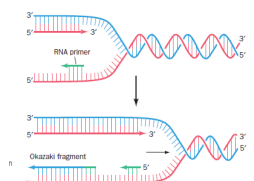
- Replication begins at one or more sites on the DNA molecule - **replication origin**
- DNA replicating enzyme DNA polymerase (two subunits/dimer) and other enzymes begin catalyzing the addition of nucleotides to the growing complementary strands of DNA
- DNA polymerase can add nucleotides only to a chain of nucleotides that is already paired with the parent strands



Cell Biology

Biomolecules & Molecular organization
DNA replication: Overview

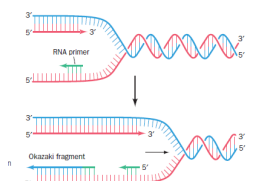
- DNA polymerase cannot link the first nucleotides in a newly synthesized strand
- A type of RNA polymerase enzyme called primase, constructs a short (10 nucleotides long) RNA strand (RNA primer) complementary to the parent DNA template



Cell Biology

Biomolecules & Molecular organization
DNA replication: Overview

- DNA polymerase adds Deoxy-ribonucleotides to the RNA primer to construct the new DNA strands
- RNA nucleotides in the primers are then replaced by DNA nucleotides

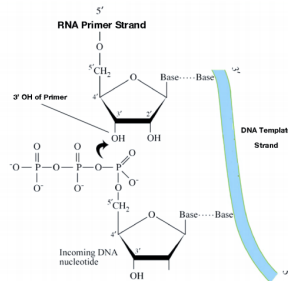


Cell Biology

Biomolecules & Molecular organization

DNA replication: Overview

- DNA polymerase can add nucleotides only to the 3' end of a DNA strand (the end with an OH group attached to a 3' carbon atom)
- Replication always proceeds in the 5' → 3' direction on a growing DNA strand

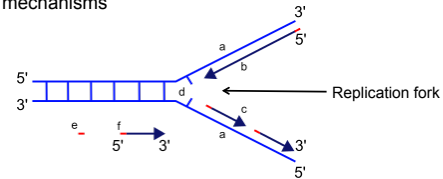


Cell Biology

Biomolecules & Molecular organization

DNA replication: Overview

- Two parent strands of a DNA molecule are antiparallel, the new strands are oriented in opposite directions along the parent templates at each replication fork
- The new strands must be elongated by different mechanisms

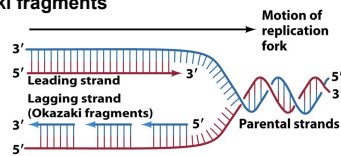


Cell Biology

Biomolecules & Molecular organization

DNA replication: Overview

- The strand which elongates towards the replication fork (**leading strand**) is built up simply by adding nucleotides continuously to its growing 3' end
- The strand that elongates away from the replication fork (**lagging strand**), is synthesized discontinuously as a series of short segments that are later connected - **Okazaki fragments**

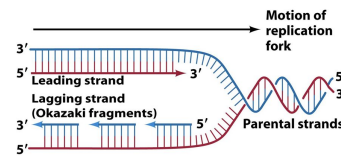


Cell Biology

Biomolecules & Molecular organization

DNA replication: Overview

- Okazaki fragments are about
 - 100 to 200 nucleotides long in eukaryotes
 - 1000 to 2000 nucleotides long in prokaryotes
- Each Okazaki fragment is synthesized by DNA polymerase in the 5' → 3' direction, beginning at the replication fork and moving away from it



Cell Biology

Biomolecules & Molecular organization

DNA replication: Overview

- The enzyme **DNA ligase**, attaches the Okazaki fragments together in the lagging strand
- DNA is further unwound, new RNA primers are constructed, and DNA polymerase then jumps ahead 1000 to 2000 nucleotides (toward the replication fork) to begin constructing another Okazaki fragment
- Replication of DNA is **semidiscontinuous**

Cell Biology

Biomolecules & Molecular organization

DNA replication: Replication process

- Consists of five interconnected steps
 - Opening up the DNA double helix
 - Building a primer
 - Assembling complementary strands
 - Removing the primer
 - Joining the Okazaki fragments

Cell Biology

Biomolecules & Molecular organization

DNA replication: Replication process

- Opening up the DNA double helix
 - DNA double helix must be opened up and its strands separated from each other for replication to occur
 - Consists of four stages
 - Stage 1: Initiating replication
The binding of **initiator proteins** to the replication origin initiates the opening of the helix
 - Stage 2: Unwinding the duplex
Enzyme **helicase** binds to the strand and moves along separating the two strands

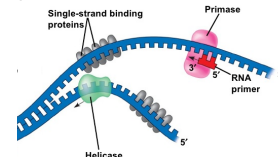
Cell Biology

Biomolecules & Molecular organization

DNA replication: Replication process

Opening up the DNA double helix

- Stage 3: Stabilizing the single strands
The unwound portion of the DNA double helix is stabilized by **single-strand binding protein**, which binds to the exposed single strands



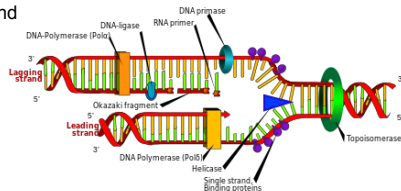
Cell Biology

Biomolecules & Molecular organization

DNA replication: Replication process

Opening up the DNA double helix

- Stage Four: Relieving the torque generated by unwinding
To relieve the resulting twisting (torque), enzyme **Topoisomerase** cuts a strand of the helix, allow it to rotate around the intact strand, and then resealed the broken strand



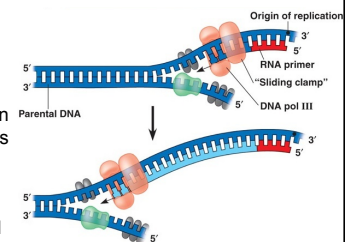
Cell Biology

Biomolecules & Molecular organization

DNA replication: Replication process

2. Building a primer

DNA polymerases require 3' primers to initiate replication
Necessary primer is a short stretch of RNA, added by a specialized **RNA polymerase** called **Primase**



Cell Biology

Biomolecules & Molecular organization

DNA replication: Replication process

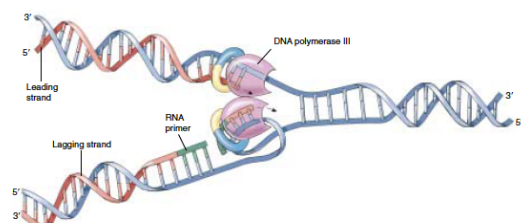
- Assembling complementary strands
 - DNA polymerase** binds to the replication fork
 - While the leading strand complexes with one half of the polymerase dimer, the lagging strand loops around and complexes with the other half of the polymerase dimer
 - The two subunits of DNA polymerase catalyzes the formation of complementary sequences on each of the two single strands at the same time

Cell Biology

Biomolecules & Molecular organization

DNA replication: Replication process

3. Assembling complementary strands



Cell Biology

Biomolecules & Molecular organization

DNA replication: Replication process

4. Removing the primer
Enzyme DNA polymerase
- now removes the RNA primer
and
- fills in the gap, as well as any gaps between Okazaki fragments

Cell Biology

Biomolecules & Molecular organization

DNA replication: Replication process

5. Joining the Okazaki fragments
Enzyme DNA ligase joins the fragments to form a continuous strand in the lagging strand

Cell Biology

Biomolecules & Molecular organization

DNA replication: Replication process

- In eukaryotes DNA is packaged in chromosomes
- Each individual zone of a chromosome replicates as a separate section called a replication unit/replicon
 - Multiple units undergo replication at any given time
 - Other than above, replication is generally similar to prokaryotic DNA replication

Cell Biology

Summary

- **DNA replication** is the biological process of producing two identical copies of DNA from one original DNA strand
- Each strand of the original DNA molecule serves as a template
- Replication begins at one or more sites on the DNA molecule
- DNA polymerase catalyze the addition of nucleotides to the growing complementary strands of DNA
- DNA polymerase can add nucleotides only to the 3' end of a DNA strand
- Replication of DNA is **semidiscontinuous**
- DNA replication consists of five interconnected steps: opening up the DNA double helix, building a RNA primer, assembling complementary strands, removing the primer, joining the Okazaki fragments